

Software Defined AIS Receiver Implementation Based on RTL-SDR and GNU Radio

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Abstract—Automatic Identification System (AIS) is a radio data system on board ships with an objective to exchange different kind of informations (static, dynamic and trip related informations). This exchange could happen between ships themselves or between ship and ground stations. While receiving these informations, AIS devices on board or ashore present in the reception area can automatically locate, identify and track AIS-equipped vessels. There are many types of AIS devices such as class A mobile stations (161.975 MHz) mandated for vessels 300GT and above, and class B mobile stations (162.025 MHz) with limited functionality and primarily used for vessels such as pleasure crafts. This paper presents the reception and decoding of AIS class A mobile station signal using RTL-SDR device with VHF antenna, GNU Radio and AIS decoder to decode the frame.

Index Terms—AIS, RTL-SDR, GNU Radio

I. INTRODUCTION

AIS was required by the International Maritime Organization (IMO) in 2004 [1]. Now, it is regarded as a safe navigational aid. It consists of exchanging data between ships, with the objective of illustrating the position of nearby boats. Due to the growth of AIS into a worldwide system that relies on a satellite constellation- Satellite based Automatic Identification System (S-AIS) [2] - since 2009. Sector participants would have been in position to maintain a close check on maritime traffic wherever it takes place. And this will give birth to new services and innovations in the field that frequently go beyond the confines of traditional maritime surveillance and decryption.

AIS stands for automatic identification system. It allows identification and tracking from other ships and from shore stations. A broadcast AIS improves safety and overcomes the

limitation of radars. The AIS transponder unit is meant to automatically broadcast the vessel's position, course and speed over ground as well as static and voyage related information over VHF channels (161.975 MHz for class A mobile stations and 162.025 MHz for class B mobile stations) with 12.5W transmission power [3]. This enables the mariners to be aware of other ships long before visual contact is established, allowing them to take required measures good time. Fig. 1 shows the data types included in AIS transmission.

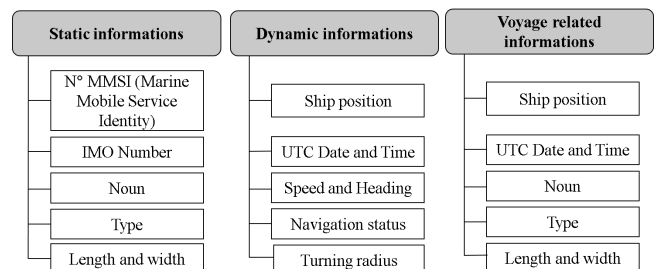


Fig. 1. AIS data type

II. BASIC THEORIES

A. Self-Organized TDMA communication protocol of AIS

The AIS transponder consists of a GPS receiver that provides position, speed, and time information to a computer. The computer then utilizes a VHF radio to share the information with other AIS stations. This method is known as Self-Organized Time Division Multiple Access (SO-TDMA) [4], and it only requires one radio channel for transmission and reception.

B. Time Slot method

The AIS transmitter uses a time slot of only few hundreds seconds to send its own data. Meanwhile it reserves another time slot for the next message. We notice that all AIS equipped ships and shore stations go through the same procedure.

C. HDLC frame standards

Everything is regarded as a sequence of bits in a Bit Oriented approach. HDLC is also a bit-oriented protocol. It is an abbreviation for High Data Link Control. This protocol is commonly used in AIS communications. The main change is that in AIS broadcasts, the address and control fields are omitted because the Data field covers these functions.

D. NMEA message standards

The systems employed in the navy are always evolving, enhancing the user's safety and comfort. The NMEA (National Marine Electronics Association) defines and controls the NMEA 0183 standard, which is a serial communication interface for conveying a "sentence" to one or more receivers. We mention that the NMEA 0183 frame protocol utilizes ASCII characters. In Fig. 2 we have the general structure of NMEA encapsulated sentence described in Table I.

TABLE I
NMEA SENTENCE DESCRIPTION

Field 0	In this field there is the sentence beginning delimiter which is "!" in the case of AIS transmissions.
Field 1	The 5-character address field is located here. The first two letters indicate the current sentence's source identification. The final three characters refer to the format of the sentence.
Field 2	Contains the overall amount of packed sentences required for the entire data.
Field 3	Contains the decimal number of the existing sentence out of the whole number of sentences.
Field 4	For multi-sentence communications, This field refer to a decimal sequential identifier that differentiates one packet message with one or more sentences from another encapsulated message with the same phrase format string.
Field 5	Represents the data holding block. Its size is controlled by the address field. This field's bits are all packaged into six-bit ASCII characters.
Field 6	Provides the decimal number of fill bits that were inserted at the end of the data if needed to form the final six-bit character.
Field 7	We have the checksum delimiter, in order to separates the NMEA phrase from the NMEA checksum field. It denotes the end of a sentence.
Field 8	Carries the NMEA checksum, which is represented by two hexadecimal characters. The checksum is computed by performing an Exclusive-Or operation across all characters in the phrase between the start "!" and stop "*" delimiters.

!aacc,X₁,X₂,X₃,c—c,X₄*hh

0 1 2 3 4 5 6 7 8

Fig. 2. General structure of NMEA encapsulated sentence

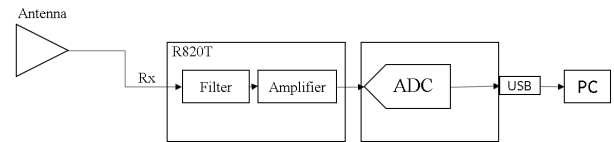


Fig. 3. Structural scheme of RTL-SDR

E. RTL-SDR

RTL-SDR is a low-cost hardware device used for Software Defined Radio real-time communication experiments [6]. It can only receive and has a maximum bandwidth of about 3MHz as well as a frequency range of 24MHz to 1766MHz. The structural layout of the employed RTL-SDR device is shown in Fig. 3.

F. Software defined Radio

A software defined radio system is a radio signal receiving and transmitting system in which the majority of the functions are provided in the digital domain [5]. It enables reconfigurability by managing program functionality. It should also be able to adapt to the system's dynamic behavior. It is a radio in which a part or all of the physical layer functions are software defined.

G. GNU Radio

To build and SDR receiver or transmitter, some appropriate drivers and software are required. GNU Radio is one of these freeware programs. It is a free software development toolkit includes different blocks for reception, transmission and signal processing [7]. These blocks can be merged to form a more complex unit named a flowgraph. To mention that these blocks are written in C++ or python, and it allows you to add additional user-written blocks as extra blocks in order to utilize them when it is necessary.

III. SOFTWARE DESIGN

A. System structural scheme

The RTL-SDR device, which is in charge of receiving signals, is attached to the laptop and setup using GNU Radio software. The RTL-SDR device receives the signal and forwards it to the GNU Radio scheme. The GNU Radio program also includes an NRZI decoding method. The bit stream that results will be stored in a binary file. HDLC to AIS NMEA converter software will be used to convert the data into an AIS NMEA message as shown in Fig. 4. To open and decode this message, the AIS decoder software will be employed.

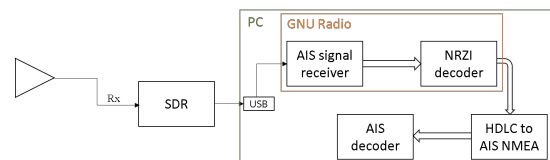


Fig. 4. Structural scheme of AIS receiver system

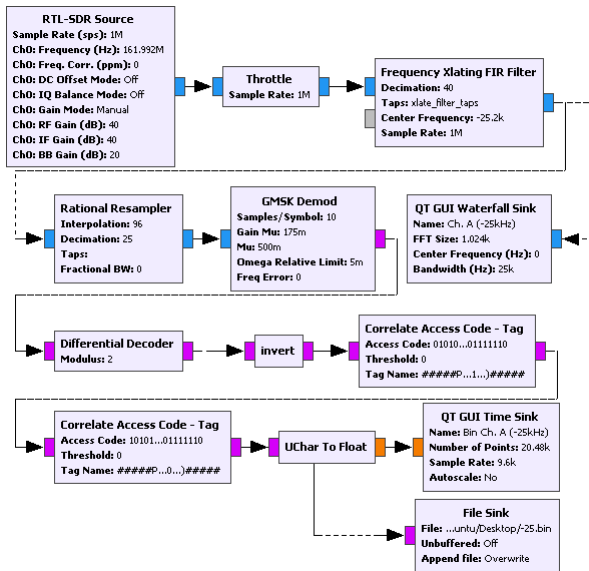


Fig. 5. AIS receiver flowgraph

B. GNU Radio flowgraph

In this work, we used the GNU Radio version 3.7.11 for UBUNTU. Fig. 5 depicts the implemented AIS receiver flowgraph. Our software only processes data from AIS channel A (161.975 MHz). The flowgraph is divided into four key sections:

- The first part is the source unit, which is the RTL-SDR receiving device.
- second part is the output unit, which is where we will save data as files.
- The third part deals with signal processing, which includes the demodulation process.
- The fourth and last section focuses on visualizing and presenting the signal processing progress at each level.
- The first block is the RTL-SDR source. It is the processing's starting point, including some required parameters. (gain = 40dB ; sample rate = 1MHz ; central frequency = 161.991MHz).
- The output unit is represented by both of the file sink's blocks. The first one on the left is connected to the source unit and acts as a recording device to preserve the received data. The one on the right followed the processing block in order to keep the processed data bit stream in a binary file.
- frequency sink block aims to show the signal's spectral density.
- Correlate access code Block looks for a set of bits that fits the acces code and highlights them visually.
- Uchar to float block convert the binary stream into float type value for further use in the time sink block.
- Time sink block shows the temporel representation of the entering value stream.
- Frequency FIR filter block removes the AIS channel frequency with a sampling rate of 25KHz. As previously

stated, various settings are required (sample rate = 1MHz ; central frequency = 162MHz ; decimation = 40).

- The rational resampler is a support block for the following GMSK demodulator block. It prepares the signal's sample rate before applying GMSK demodulation on it.
- The signal demodulation procedure is provided by the GMSK demodulation block. For current transmission, an input signal with a specified sample rate is required (9.6 KHz in AIS transmission case).
- The NRZI decoding process is referred to by the differential decoder and invert blocks. According to the NRZI decoding standard, the Differential Decoder unit is the Exclusive-Or operation for every pair of neighbor bits in the stream. Meanwhile the Invert block performs bit inversion on every bit in the stream.

IV. RESULTS

Fig. 6 shows a binary signal representing the NRZI decoded bit sequence. This figure refers to the bit sequence seen below: "010101010101010101010101111100011100000110110101001011100001101101011001000001000110011011101010000110001100100100010110000011010000100111000010111100110111010111000111100000100101101101110111001101110111110"

The preamble sequence (010101010101010101010101) is included in the message frame. Which appear at the start of the data sentence. In addition, the data frame contains the frame payload as well as the HDLC start and stop flags (01111110), which identify the start and end of the sentence. The bit stream is retained as a binary file and will be processed later by the AIS NMEA message generator Fig. 7.

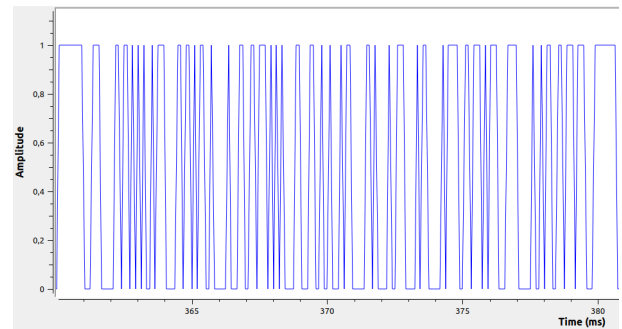


Fig. 6. NRZI decoded frame

```
System ::      ::Channel:      < UNDEFINED. >
          ::File :      < UNDEFINED. >

> Menu ::      ::Options:

          { file } to define file name and type.
          { chnl } to change AIS channel.
          { --- } to generate AIS Messages. [Define file first]
          { exit } to exit.

::
```

Fig. 7. AIS NMEA message generator

There are four options in the main menu: "exit" to quit the application, "Chnl" to specify the AIS Channel A. The "file" option enables you to provide the source file, and the "gen" command starts the primary data processing step. The operation begins with the source file being opened and all previously detected relevant data being copied to the buffer. This will come to an end when the HDLC stop flag sequence (01111110) is encountered. In the subsequent phases, all unneeded bits will be deleted which include the HDLC start and stop flag sequence and the preamble sequence. As well as the last 16 bits preceding the stop flag sequence indicating the received check sequence. The remaining data sequence contains the appropriate AIS data bits. As the last binary alteration of the processed data, bit flipping is performed on the appropriate AIS data bits. It involves combining bits into groups of eight bits and then flipping them so that the first bit in the group becomes the last one and vice versa. The resulting data sequence contains accurate AIS message data and may now be decrypted based on AIS message structure standards: "00000100011011001001010100001110010110110000000000011001110110000101010001100010010011010001011000110010001010011101000010110011110101100010000100100011001101". Table. II shows the decoded segments of the resulting data sequence with their meaning. In the AIS NMEA message, the data bits are grouped by 6 and each group is turned into a 6-bit ASCII symbol as shown in Fig. 8. Table .III describes the fields of the final NMEA message generated.

! AIVDM, 1, 1, , A,
 16yE3ls03Ow9j4kzWMin`WfWIS3N,
 0*7A

Fig. 8. AIS NMEA message

V. DISCUSSION

The primary purpose of this study was to create open maritime AIS on software defined radio, that will be capable of using a wide range of SDR devices. It was built around an RTL-SDR device AIS receiver with various drawbacks for channel A mobile stations. The system necessitates independent recording of input signal with demodulation and initial NRZI decoding via the GNURadio receiver flowgraph, as well as major binary data treatment via NMEA message creation and minimum data decoding via the HDLC to AIS NMEA converter. However, the entire decoding cycle takes only a few minutes, and also the price of this receiver seems to be much lower than that of the specialized device. Also Developments in ASIC, FPGA, SoC, open-source software have paved the way for modern marine electronics innovation. whether it's MFDs, depth sounders, thermal cameras, or AIS Transponders. A significant amount of signal processing needed to encode and decode AIS communications may now be specified in

TABLE II
NMEA SENTENCE DESCRIPTION

Bits	Parameter	Binary segment value	Decimal form	Decoded value and meaning
1-6	Message type	000001	1	Class A AIS position report
7-8	Repeat indicator	00	0	Default repeat settings
9-38	Source MMSI number	0110110010010101000011100101	455426966	First 3 digits refers to the MID number (455-Maldivia)
39-42	Navigation- al status	1100	12	power-driven vessel pushing ahead
43-50	Rate of turn	000000000	0	0 degrees/min
51-60	Speed over ground	0011001110	206	20.6 knots
61	Position accuracy	1	1	High (10m)
62-89	Longitude	100001010101001100010010010	139794578	56.445 degree East
90-116	Latitude	011010001011001100100010	54889610	53.456 degree North
117-128	Course over ground	011110100001	1953	195.3 degrees
129-137	True heading	011001111	207	207 degrees
138-143	Time stamp	101011	43	43 second
144-145	Special maneuver indicator	00	0	not available
146-148	Spare	010	2	not used
149	RAIM-Flag	0	0	RAIM not in use
150-168	Communication state	00100100000110101	73933	Stack of communication state specifications

TABLE III
NMEA SENTENCE FIELDS DESCRIPTION

Fields 0;1	"!AIVDM" indicates the beginning of a VDM format sentence (received from AIS station).
Field 2	"1" denotes the number of sentences required to accommodate all sent data.
Fiel 3	"1" denotes the current sentence's sequential number.
Field 4	It is the sequential identifier for multi sentence messages Since it is one sentence message, this field is empty
Field 5	"A" refers to the receiving channel (161.975MHZ)
Field 6	Represents the received and processed AIS data. It contains 168 bits of information.
Field 7	"0" means that 0 last bits in payload are fill-bits and therefore must be ignored.
Field 8;9	determines the end of the sentence and returns the hexadecimal NMEA checksum generated prior to transmission to check for probable error appearance.

software. This substitutes a major amount of the more typical hardware-intensive analog solution, making it easier to create, quicker, and more capable.

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